

PROCESS SHEET

Project: ACE Simple Tank E2

Language: English

DWG.#: 0000-0009

Client: Armored Combat Extended



R-0-1-XIce Frost

GMOD Engineer Academy
The Machinist's Sanctuary





Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
10	FINDING MAP'S NORTH DIRECTION		
A	2 METHODS FINDING NORTH		
a	USE THE PAINT TOOL TO FIND THE NORTH DIRECTION OF THE MAP (REFER TO PICTURE #10a)	PAINT → SMILE	PLACE THE SMILE ON THE GROUND, THE UPWARD SIDE POINTS NORTH (THE WAY IT FACES IS NORTH)
b	USE YOUR OWN SPRAY (PIC #10b)	FIND YOUR BIND KEY TO USE YOUR OWN SPRAY (BIND KEY T IN MY CASE)	USE THIS METHOD AFTER REMEMBERING THAT THE TOP OF YOUR SPRAY POINTS NORTH.
20	BUILDING SIMPLE TANK PART 1		
A	CONSTRUCT BASIC GEOMETRY SHAPE WITH PRECISION ALIGENMENT (AKA PA)	PRECISION ALIGENMENT (PA) & SMARTSNAP	-----N/A-----
a	SPAWN PROP (BASEPLATE): RECT 78x96x1.5	SProps → PLATES THIN (Q MENU)	MAKE SURE YOU'RE FACING MAP'S NORTH DIRECTION (PIC #20a)
b	CHECK FOR SQUARENESS (PIC #20b) DEMO: VIDEO LINK	G-RULE TOOL – ADVANCED MEASUREMENT	- HAMMER UNIT (UNIT) - BASIC (SNAP TO PROP)
25	PRECISION ALIGENMENT SETTINGS	PRECISION ALIGNMENT RETOOLED (MODERN 2025) BY ACF-3 TEAM OR PRECISION ALIGNMENT (OLD 2015) BY ./CHOSEN	- SMARTSNAP (DEFAULT SETTINGS) - POINT – HITPOS - ENABLE CONSTRUCT DISPLAYS - SNAP TO EDGES - SNAP TO CENTRE LINES (IF REQUIRED) - SNAP SENSITIVITY: 100
A	TERMS I'LL BE USING: - HIGHEST POINT (PIC# 25a) - MIDPOINT (PIC #25b) - LOWEST POINT (PIC #25c)		DEMO: VIDEO LINK 1 (SMART WAY) DEMO: VIDEO LINK 2 (OLD SCHOOL WAY)

Picture Board



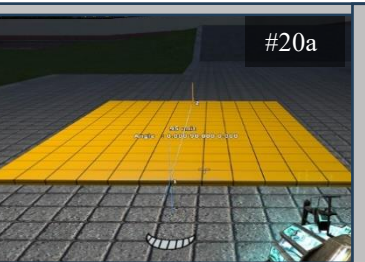
#10a

UPWARD SMILE POINTS NORTH



#10b

TOP OF YOUR SPRAY POINTS NORTH



#20a

BASE PLATE (NORTH DIRECTION)



#20b

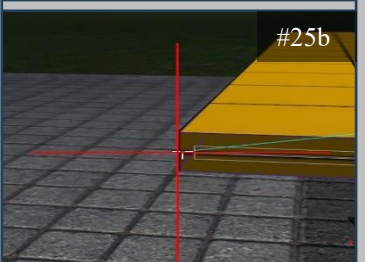
95 unit
Angle : -0.000 90.000 0.000

100% Square Verified with G-Rule Tool - Advanced Measurement



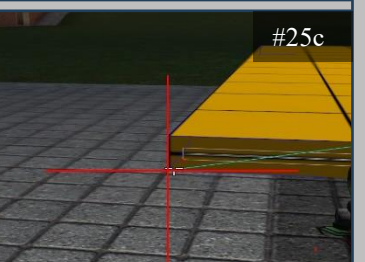
#25a

HIGHEST POINT



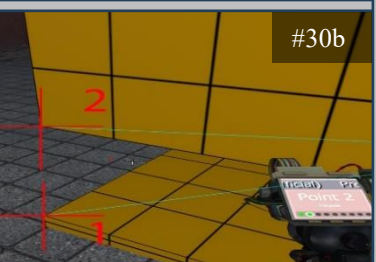
#25b

MIDPOINT



#25c

LOWEST POINT



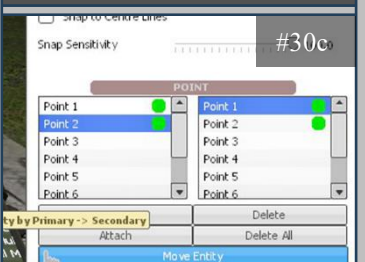
#30b

FRONT VIEW



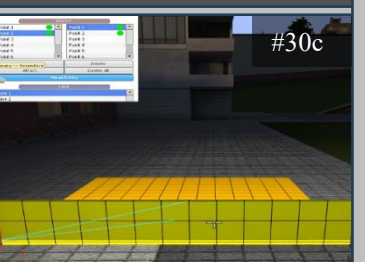
#30c

SELECT ENTITY (RIGHT CLICK)



#30c

MOVE ENTITY



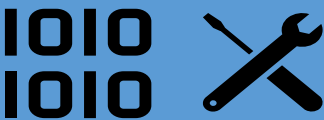
#30c

THE RESULT



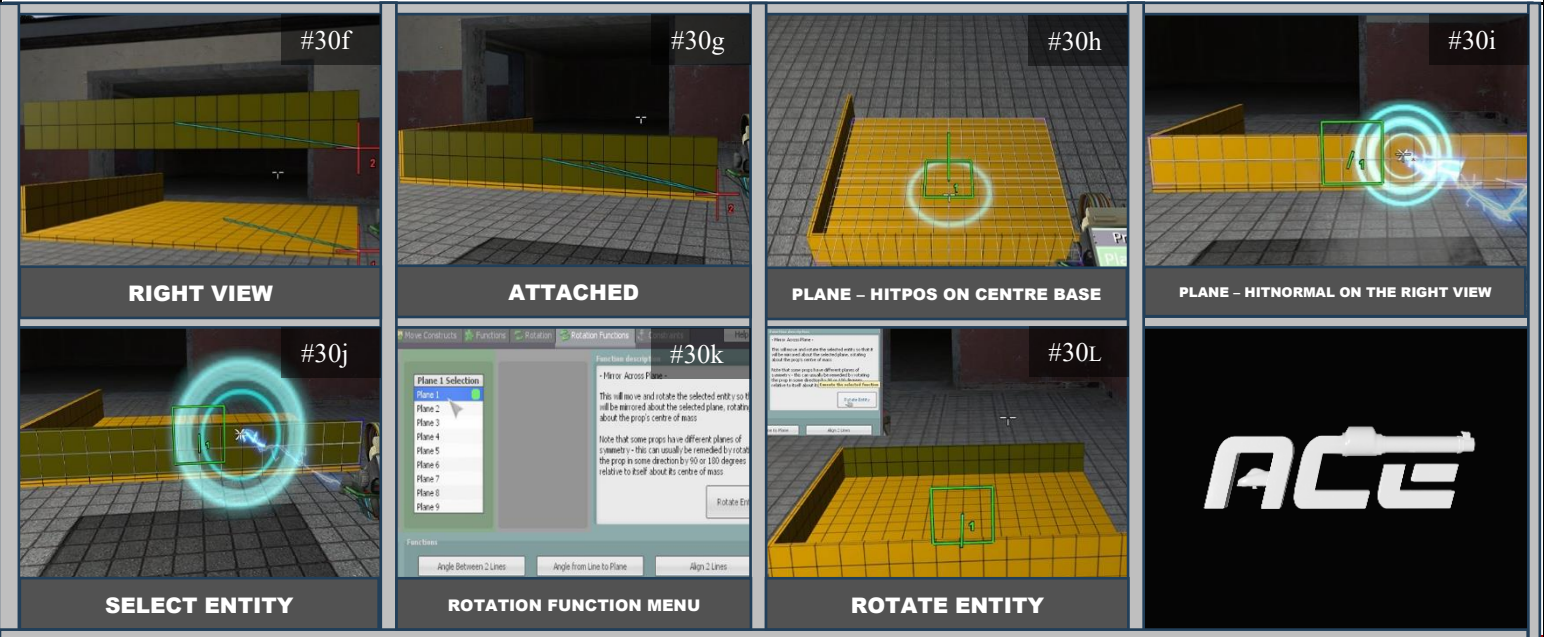
ACE

Enjoy Wiremod, Starfall, or ACF/ACE? Become a contributor and help these addons keep evolving!



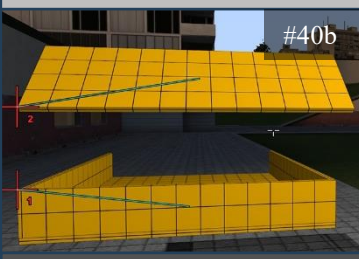
Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
30	BUILDING SIMPLE TANK PART 1 (CONTINUED)		
A	CONSTRUCT BASIC GEOMETRY SHAPE WITH PRECISION ALIGENMENT (AKA PA)		
a	SPAWN PROP (FRONT VIEW): SPROP 12x78x1.5 	SPROPS → PLATES THIN (Q MENU)	
b	GENERATE 1ST POINT (HIGHEST) ON BASEPLATE (PIC #30b) GENERATE 2ND POINT (LOWEST) ON 12x78x1.5 (PIC #30b)	- PA TOOL - LEFT CLICK TO GENERATE 1ST POINT - HOLD SHIFT + LEFT CLICK TO GENERATE 2ND POINT	POINT – HITPOS PA SETTINGS (REFER TO PROCEDURE #25)
c	TO ATTACH RIGHT CLICK ON 12x78x1.5 & MOVE ENTITY (PICS #30c)	- RIGHT CLICK - HOLD C - CLICK MOVE ENTITY	DEMO: VIDEO LINK (ACTION 30a TO 30d)
d	RIGHT CLICK ON THE GROUND & DELETE ALL POINTS (MAKE THIS HABIT)	HOLD C & DELETE ALL (POINT SECTION)	TO AVOID CONFUSION IF YOU'RE NEW TO PA TOOL BEFORE ATTACHING ANOTHER PROP!
e	SPAWN PROP (RIGHT VIEW): SPROP 12x96x1.5 	SPROPS → PLATES THIN (Q MENU)	
f	GENERATE 1ST POINT (HIGHEST) ON BASEPLATE (PIC #30f) GENERATE 2ND POINT (LOWEST) ON 12x96x1.5 (PIC #30f)	PA TOOL	
g	ATTACH THE PROP (PIC #30g) (SIMILAR PROCEDURE #30c)		
B	MIRROR THE PROP ACROSS THE PLANE (COPYING ITS SYMMETRY TO THE OTHER SIDE)		PLANE - HITPOS + HITNORMAL PA SETTINGS (REFER TO OP #25 FOR SETTINGS)
h	HITPOS ON THE CENTRE BASEPLATE	PA TOOL	PLANE – HITPOS + HITNORMAL (PIC #30h)
i	HITNORMAL ON 12x96x1.5 YOU WANT TO MIRROR ACROSS (LEFT CLICK)	-----	PLANE – HITNORMAL (PIC #30i)
j	SELECT ENTITY (RIGHT CLICK)	-----	(PIC #30j)
k	PRESS R TO OPEN THE MENU	ROTATION FUNCTIONS → MIRROR ACROSS PLANE	TO SEE PLANE 1 SELECTION LEFT CLICK MIRROR ACROSS PLANE
L	LEFT CLICK #PLANE 1 TO MIRROR (PIC #30K & PIC 30L)	HOLD SHIFT + LEFT CLICK → ROTATE ENTITY	DEMO: VIDEO LINK (ACTION 30h TO 30L)

Picture Board



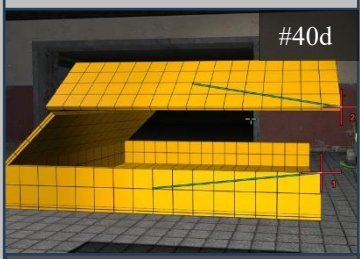
Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
40	BUILDING SIMPLE TANK PART 2		
A	CONSTRUCT BASIC GEOMETRY SHAPE WITH PRECISION ALIGNMENT (AKA PA)		
a	CLEAR ALL PRIOR PA POINTS BEFORE USING THE PA TOOL AGAIN (MAKE THIS HABIT!)	HOLD C & DELETE ALL (POINT SECTION) & (PLANE SECTION)	TO AVOID CONFUSION IF YOU'RE NEW TO THE PA TOOL— UNLESS YOU'RE GMOD SEASONED BUILDER
b	SPAWN PROP (FRONT VIEW): SPROP 24x78x1.5 & TILT 45° 	SProps → PLATES THIN (Q MENU)	REFER TO PIC #40b
c	ATTACHED THE PROP WITH PA TOOL: GENERATE 1ST POINT (TOP LEFT EDGE) ON 12x78x1.5 (PIC #40b) GENERATE 2ND POINT (HIGHEST) ON 24x78x1.5 (PIC #40b)	PA TOOL	POINT – HITPOS SAME PA SETTINGS (REFER TO PROCEDURE #25)
d	SPAWN PROP (RIGHT VIEW): SPROP 24x96Ax1.5 & TILT 45° 	-----	REFER TO PIC #40d
e	ATTACHED THE PROP WITH PA TOOL: GENERATE 1ST POINT (TOP RIGHT EDGE) ON 12x96x1.5 (PIC #40d) GENERATE 2ND POINT (HIGHEST) ON 24x96x1.5 (PIC #40d)	-----	-----
f	MIRROR ON 24x96x1.5 (PIC #40f)	-----	PLANE - HITPOS + HITNORMAL • PA SETTINGS (REFER TO PROCEDURE #30B)
g	DEMO: VIDEO LINK (ACTION 40b TO 40g)	-----	WE’LL TRIM THE PARTS LATER WITH THE PROPER CLIPPING TOOL
h	CONFIRM INTERIOR IS FULLY FLUSH	-----	OPTIONAL— FINE IF IT’S NOT FLUSH FIRST TRY...
i	SPAWN PROP (TOP VIEW): SPROP 48x78x1.5 	-----	-----
j	ATTACHED THE PROP WITH PA TOOL: GENERATE 1ST POINT (MIDPOINT) ON 24x78x1.5 (PIC #40i) GENERATE 2ND POINT (MIDPOINT) ON 48x78x1.5 (PIC #40i)	-----	-----
k	APPLY WITH PROPER CLIPPING TOOL Demo: VIDEO LINK (pics #40k)	PROPER CLIPPING TOOL	SETTINGS IN THE VIDEO

Picture Board



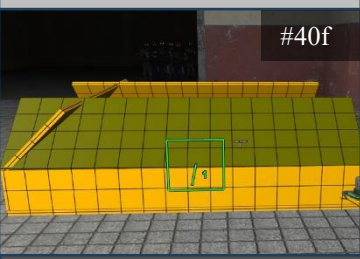
#40b

FRONT VIEW (ATTACHED)



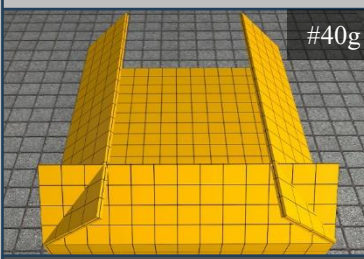
#40d

RIGHT VIEW (ATTACHED)



#40f

RIGHT VIEW (MIRROR)



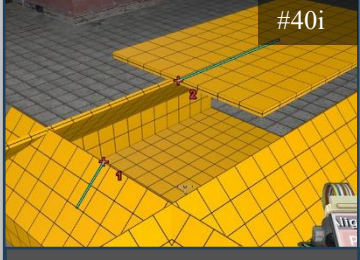
#40g

THE RESULT (Trim Parts Later)



#40h

Fully Flush On the Inside



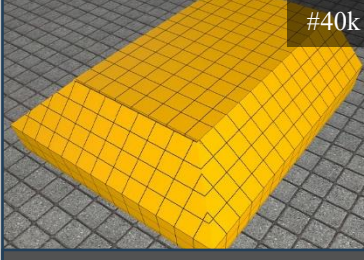
#40i

TOP VIEW (ATTACHED)



#40k

Before Proper Clipping Applied



#40k

After Proper Clipping Applied



Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
50	BUILDING SIMPLE TANK PART 3		
A	SETTING UP THE WHEELS PART 1	-----	
a	SPAWN WHEEL PROP: SPROP T_20 (WHEEL A) 	SProps → TRANSPORTATION → WHEEL A (TYPES)	-----
b	GENERATE 1ST POINT (RIGHT CENTRE) ON 12x96x1.5 (PIC #50b) GENERATE 2ND POINT (CENTRE) ON T_20 WHEEL (PIC #50c)	PA & SMARTSNAP	• SNAP TO EDGES (OFF) • SNAP TO CENTRE LINES (OFF) • MIN DISTANCE BETWEEN: 6 OR 2 (SMARTSNAP SETTINGS) • HOLD E TO SNAP
c	ATTACHED (PIC #50d) DEMO: VIDEO LINK (ACTION 50b TO 50c)		
B	STACKER TOOL	STACKER TOOL - IMPROVED	• STACK RELATIVE TO: PROP • WELD STACKED PROPS (OFF!) (PIC #50e)
d	FIND WHICH DIRECTION THE WHEEL IS POINTING (3D AXIS DIRECTION)	STACKER TOOL - IMPROVED	STACK DIRECTION: (FRONT? BACK?) MIND THE 3D CARTESIAN DIRECTION IT FACES (PIC #50f)
e	MAKE COPIES OFFSET TO THE LEFT (PIC #50e -50e)	-----	X (-BACK, +FRONT): 7 (IN MY CASE) STACK DIRECTION: FRONT (IN MY CASE)
f	DEMO: VIDEO LINK	-----	(ACTION 50e -50e)
g	-----	-----	-----
h	MIRROR THE WHEELS (4 TIMES)	PA TOOL	PLANE - HITPOS + HITNORMAL • PA SETTINGS (REFER TO OP #25 FOR SETTINGS & #30B SIMILAR PROCEDURE)
i	DEMO: VIDEO LINK		(ACTION 50h -50i)
C	APPLYING MUTI PARENT TOOL		
j	MAKE BACKUP COPY BUILD (ADV2) (BACKUP YOUR BUILD EVERY NOW AND THEN)	ADVANCED 2 DUPLICATOR HOLD SHIFT + RIGHT CLICK	DEFAULT SETTINGS
k	PULL OUT MUTI PARENT TOOL	MUTI PARENT TOOL	• ALL SETTINGS UNCHECK
L	SELECT ALL PROPS EXCEPT WHEELS & BASEPLATE	LEFT CLICK	-----
m	APPLIED	RIGHT CLICK	-----
-----	-----	-----	-----

Picture Board



#50b

RIGHT VIEW



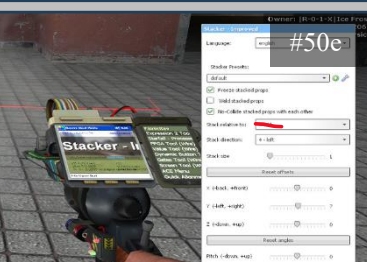
#50c

POINT #2 BACK WHEEL



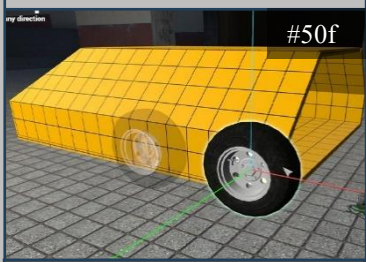
#50d

ATTACHED



#50e

STACKER TOOL



#50f

Copying Wheels 4x



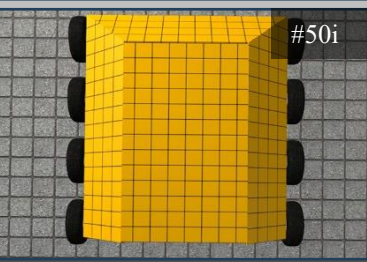
#50g

The Result



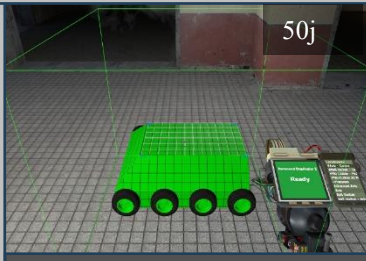
#50h

Mirror the wheels



#50i

TOP VIEW (The Result)



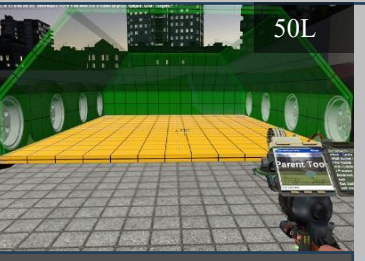
50j

Area Copy Adv2 (Backup)



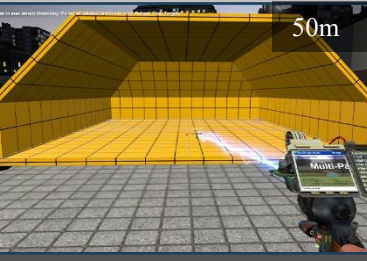
50k

MUTI PARENT TOOL



50L

Select Props (Except Base)



50m

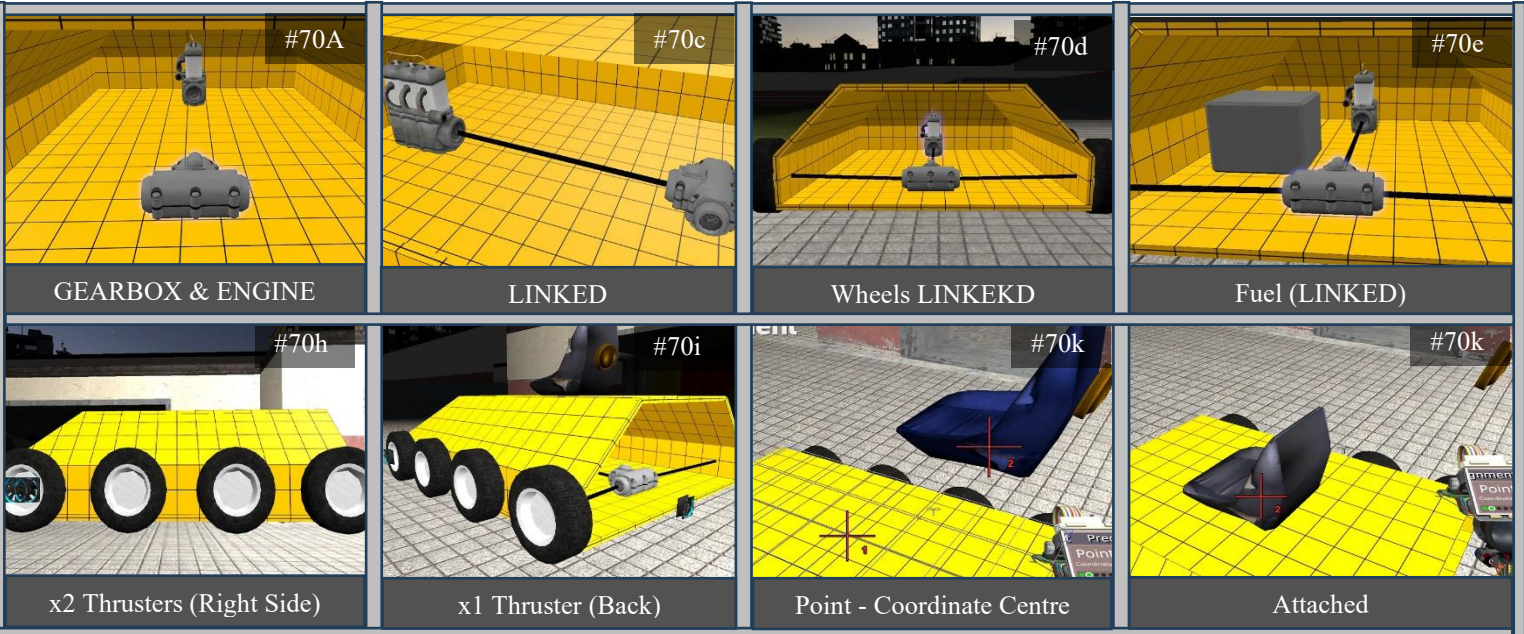
Apply



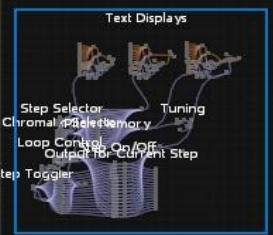
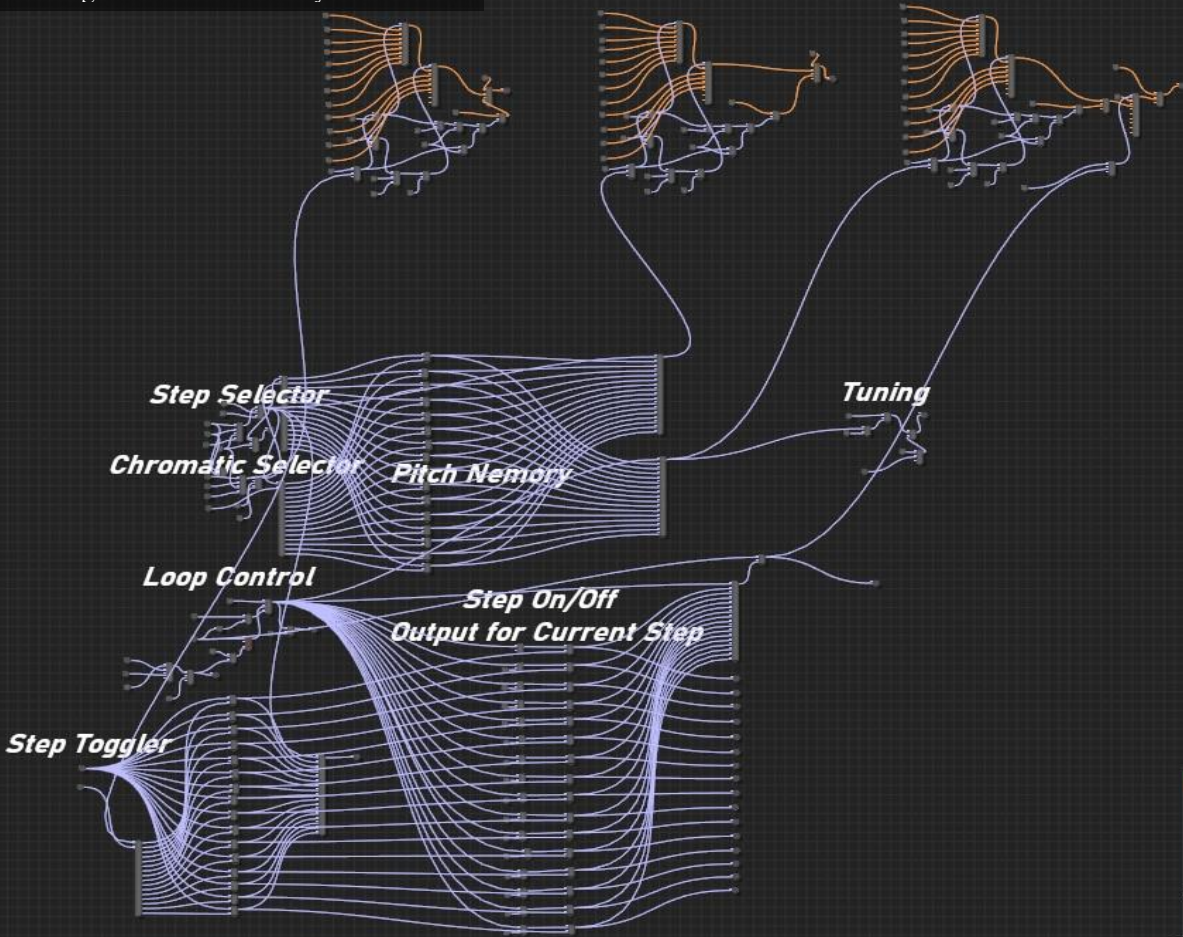
Early Build with Wiremod Gate Logics
- April 26, 2025

Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
70	BUILDING SIMPLE TANK PART 5		
A	ACE COMPONENTS SETUP PART 1		DON'T FORGET TO WELD IT!
a	SPAWN I3 ENGINE (1.1L) DIESEL 	ACE MENU (TOOL NAME)	E.G MOBILITY → ENGINES → I3
b	SPAWN DIFFERENTIAL, SMALL 	-----	SPECS IN THE DRAWING
c	LINK ENGINE TO GEARBOX	-----	RIGHT CLICK ENGINE TO GEARBOX (PIC #70c)
d	LINK GEARBOX TO REAR WHEELS	-----	RIGHT CLICK ENGINE TO WHEELS
e	SPAWN FUEL TANK (PIC #70e) DIESEL	-----	SPECS IN THE DRAWING
f	LINK FUEL TO ENGINE (PIC #70e)	-----	RIGHT CLICK FUEL TO ENGINE
g	DEMO: VIDEO LINK	-----	ACTION 70A – 70e
B	WIREMOD COMPONENTS SETUP PART 1		
h	2 THRUSTERS ON ONE WHEEL FOR TURNING		SPECS IN THE DRAWING
i	THRUSTER REAR BASEPLATE: (PIC #70i)		SPECS IN THE DRAWING
j	SPAWN CAR SEAT & ATTACH POD CONTROLLER TO IT (REFER TO PICS #70i, 70k)	*Q MENU (VEHICLES) → CHAIRS *Q MENU (WIRE) → POD CONTROLLER *RIGHT CLICK POD & RIGHT CLICK TO SEAT TO LINK (POD CONTROLLER IN HAND)	LINK POD CONTROLLER TO SEAT
k	ATTACH CAR SEAT ON TOP, AS CLOSE TO THE CENTRE AS POSSIBLE	PA TOOL	- POINT - COORDINATE CENTRE - ENABLE CONSTRUCT DISPLAYS (ON) - SNAP TO EDGES (ON) - SNAP TO CENTRE LINES (ON) - SNAP SENSITIVITY (ON)
L	- WELD CAR SEAT TO BASEPLATE - UNFREEZE POD CONTROLLER	WELD TOOL PHYSGUN	ONLY WELD CAR SEAT TO BASEPLATE — PARENTED PARTS WON'T ATTACH LEFT CLICK POD CONTROLLER WITH PHYSGUN TO UNFREEZE
m	DEMO: VIDEO LINK		ACTION 70j – 70L
n	SPAWN MY MAIN ENGINE SIMPLE TANK E2 CHIP AND PLACE ON BASEPLATE	EXPRESSION 2 TOOL	* PLACE ANYWHERE YOU LIKE * GRAB MY E2 CHIP “MAIN ENGINE SIMPLE TANK E2”
-----	-----	-----	-----
SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE

Picture Board

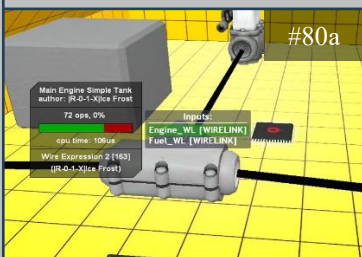


Squash's FPGA Sequencer (Workshop Dupe)
* Field-Programmable Gate Array



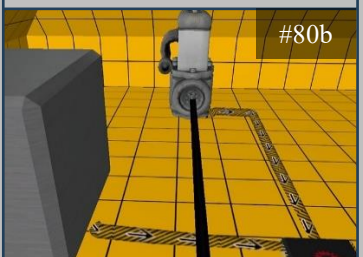
Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
80	WIRING OPERATIONS PART 1 (E2)		
A	WIREMOD & ACE WIRING SETUP P1 (E2)	WIRING TOOL	
a	ENGINE_WL TO ACE ENGINE WIRELINK [WIRELINK] (PIC #80a, #80b)	-----	WITH WIRELINK EVERYTHING IS PLUG AND PLAY
b	FUEL_WL TO ACE FUEL TANK WIRELINK [WIRELINK] (PIC #80a, #80b)	-----	-----
c	THRUSTERS (A) → TURN (E2 MAIN ENGINE OUTPUT) (PIC #80c)	-----	-----
e	BACK THRUSTER (A) → REVERSE (PIC #80c)	-----	-----
d	GEARBOX (BRAKE) TO POD CONTROLLER (SPACE)	-----	ACTIVATES BRAKE WHEN SPACE IS PRESSED/HELD
f	OPTIONAL: DISPLAY SPEED ON WIREMOD SCREEN	SEARCH SCREEN IN WIREMOD (Q MENU)	SPEED DISPLAY SCREEN (A) → THROTTLE_OUTPUT (E2)
90	SETTING UP GIMBAL & ACE GUNS & AMMO BOXES PART 1	-----	IF THE CAR SEAT IS NOT PLACED YET, REFER TO #70j OP
A	SETTING UP GIMBAL (FACER) PART 1		MAKE SURE YOU HAVE MODEL PROPERTY ADDON INSTALLED!
a	SPAWN GIMBAL (FACER)	WIREMOD (Q MENU)	PICK ANY MODEL
b	SPAWN CYLINDER_12x42' OR PICK YOUR CHOICE 	SPROPS → CYLINDER (Q MENU)	COPY MODEL HOLD C + RIGHT CLICK ON CYLINDER MODEL
c	CHANGE GIMBAL'S MODEL (REFER TO PIC #90c)	HOLD C + RIGHT CLICK ON GIMBAL'S MODEL	SET MODEL: PASTE THE MODEL COPIED EARLIER
d	IF GIMBAL STUCK UNDER THE MAP PULL IT UP WITH PHYSGUN (PIC #90d)	-----	-----
e	STRAIGHTEN THE GIMBAL WITH PHYSGUN (REFER TO PICS #90e)	HOLD E + SHIFT TO SNAP THE GIMBAL TO 45° ANGLES WHILE MOVING THE MOUSE	-----
f	DEMO: VIDEO LINK	-----	ACTION 90a – 90e
-----	-----	-----	-----
-----	-----	-----	-----
-----	-----	-----	-----
SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE

Picture Board



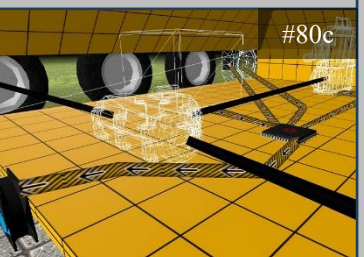
#80a

Wiring E2 (Main Engine)



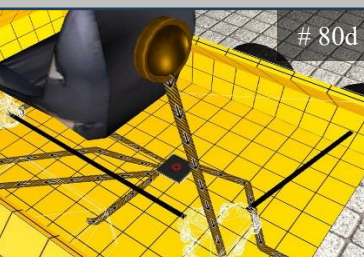
#80b

EXAGGERATED WIRES #1



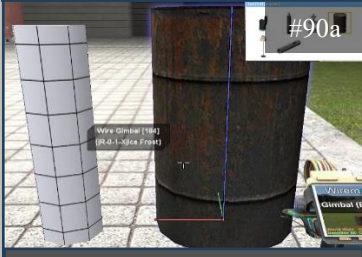
#80c

EXAGGERATED WIRES #2



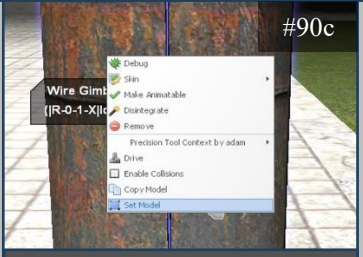
#80d

Brake Enabled



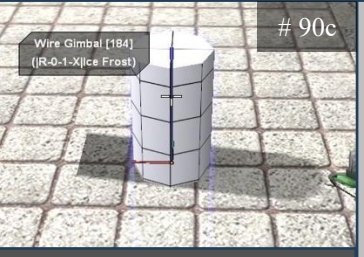
#90a

Replacing Gimbal Model



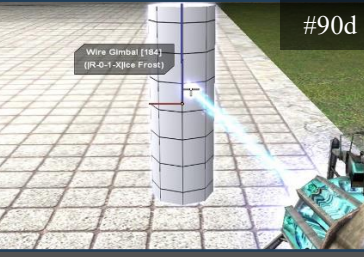
#90c

Before



#90e

After



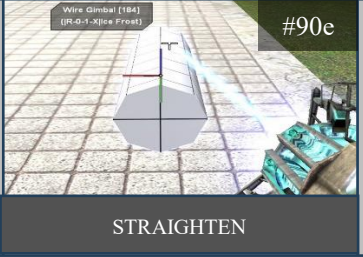
#90d

Pull up



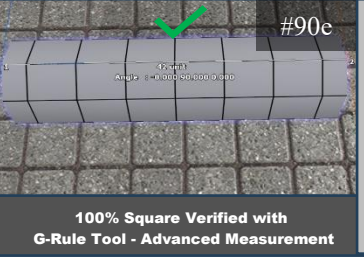
#90e

Snap to 45° Deg



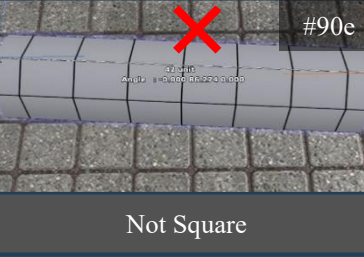
#90e

STRAIGHTEN



#90e

100% Square Verified with G-Rule Tool - Advanced Measurement



#90e

Not Square

Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
100	SETTING UP GIMBAL & ACE GUNS & AMMO BOXES PART 2		
A	SETTING UP GIMBAL (FACER) PART 2 (FINAL)		
a	GENERATE 1ST POINT TOP BASEPLATE (REFER TO PIC #100a)	PA TOOL	POINT – COORDINATE CENTRE PA DEFAULT SETTINGS
b	GENERATE 2ND POINT GIMBAL (REFER TO PIC #100b)	HOLD SHIFT + LEFT CLICK TO GENERATE 2ND POINT	
c	SELECT ENTITY (RIGHT CLICK)		
d	MOVE ENTITY (HOLD C TO OPEN MENU)		
e	FIND THE GIMBAL’S FACING DIRECTIONS AND THE DIRECTION TO RAISE IT (3D AXIS)	STACKER TOOL - IMPROVED	STACK DIRECTION: (RIGHT? FRONT?) MIND THE 3D CARTESIAN DIRECTION IT FACES (PICS #100e), in my case is “right direction”
f	HOLD C & RIGHT CLICK ON GIMBAL TO BRING UP PRECISION TOOL CONTEXT MENU		CHOOSE HOW HIGH TO RAISE THE GIMBAL: RIGHT → -50 UNITS OR LEFT → +50 UNITS
g	DEMO: VIDEO LINK		ACTION 100a – 100f
h	PARENT GIMBAL (LEFT CLICK) → RIGHT CLICK ON BASEPLATE (PIC #100h)	MULT - PARENT TOOL	ALL SETTINGS UNCHECK
110	SETTING UP GIMBAL & ACE GUNS & AMMO BOXES PART 3		
A	SETTING UP ACE GUNS	ACE MENU (TOOL)	GUNS → CANNON → 37MM
a	STRAIGHTEN THE 37MM CANNON WITH PHYSGUN (REFER TO PICS 110a)	HOLD E + SHIFT TO SNAP THE CANNON TO 45° ANGLES WHILE MOVING THE MOUSE	
b	GENERATE 1ST POINT 37MM CANNON (REFER TO PIC #110b) GENERATE 2ND POINT GIMBAL (REFER TO PIC #110b)	PA TOOL	POINT – COORDINATE CENTRE PA DEFAULT SETTINGS

Picture Board

#100a

1st Point

#100b

2nd Point

#100c

Select Entity

#100d

Move Entity (Centre)

#100e

Raising Gimbal After Centering

#100f

PRECISION TOOL CONTEXT

#100e

3D CARTESIAN

#100h

Parent Gimbal

#110a

100% Square Verified with
G-Rule Tool - Advanced Measurement

#110b

Centre to Centre

#110c

Move Gun Out Slightly & Parent it

Early Build with FPGA
- September 18, 2025

ACE Simple Tank FPGA
cpu time: 161us
peak cpu time: 293us
Wire FPGA (149)
[R-0-1-X]Ice Frost

Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
c	MOVE GUN OUT SLIGHTLY (SIMILAR PROCEDURE #100e) (PIC #110c)	PRECISION TOOL CONTEXT MENU	-----
d	PARENT 37MM CANNON (LEFT CLICK) → RIGHT CLICK ON GIMBAL (PIC #110c)	MULT - PARENT TOOL	THIS STEP IS VERY IMPORTANT! ALL SETTINGS UNCHECK
B	ADDING 2ND ACE GUN (PIC #110e)	HEAVY MACHINE GUN → 20MM	ENSURE SUFFICIENT CLEARANCE TO PREVENT GUNS FROM FIRING INTO EACH OTHER!
e	FOR THE 2ND ACE GUN, FIGURE OUT HOW TO ATTACH IT PRECISELY TO THE GIMBAL. ANSWER IN THE DEMO VIDEO (MY OWN METHOD)	-----	GOOD LUCK!
f	DEMO: VIDEO LINK	-----	ACTION 110a – 110e
g	OPTIONAL: CHANGE MUZZLES, GUN MODELS, SKIN & RELOAD PISTONS (PIC #110e)	HOLD C, RIGHT CLICK ON ACE GUN YOU WANT TO MODIFY MODEL → BODY GROUPS OR → SKIN (IF AVAILABLE)	FUN FACT: YOU CAN CUSTOMIZE APPEARANCE OF CERTAIN ACE GUNS!
120	SETTING UP GIMBAL & ACE GUNS & AMMO BOXES PART 4	-----	-----
A	SETTING UP AMMO BOXES	-----	-----
a	SPAWN AMMO BOXES AND PLACE THEM INSIDE THE TANK INTERIOR	-----	SPECS IN THE DRAWING
b	PARENT AMMO BOXES & APPLY IT TO THE BASEPLATE	MULT - PARENT TOOL	ALL SETTINGS UNCHECK
c	LINK AMMO BOXES BASED ON ACE GUN TYPE (PICS #120c)	ACE MENU (TOOL) (RIGHT CLICK)	E.G 37MM CANNON AMMO BOX → 37MM CANNON GUN
d	SPAWN MY MOUSE AIM E2 CHIP ONCE, LEFT IT ON THE GROUND, AND NEVER TOUCHED IT AGAIN!	EXPRESSION 2 TOOL	* GRAB MY E2 CHIP “MOUSE AIM BASIC V1”
-----	-----	-----	-----
-----	-----	-----	-----
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SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE	SEE NEXT PAGE

Picture Board

#110c

End Result

#110e

Customizable Appearance

#120a

Ammo Boxes

#120b

Parent Ammo Boxes

#120c

Link Ammo Boxes to ACE Guns

#120c

“Link Successful” Pop Up

#120d

Mouse Aim Basic V1 E2

ACE

ACE Simple Tank FPGA Chip (Inside)

Throttle Logic

Throttle W

Throttle S

Automatic Starter Logic

Thrust 1 Logic for turning

Mouse Aim Logic

Prepared by: [R-0-1-X]Ice Frost

Date: October 25, 2025 – August 23, 2026

Page 9 of 10

Routines, Sub-Routines, Operations	Description and Remarks	Tools/Menu/list/controls	Remarks #2/ Settings
130	WIRING OPERATIONS PART 2 (E2)	WIRING TOOL	-----
A	WIREMOD & ACE WIRING SETUP P2 (E2)	-----	-----
a	GIMBAL[WIRELINK] TO GIMBAL [WIRELINK] (PIC #80a)	-----	MOUSE AIM BASIC V1 E2 CHIP HOLD RIGHT CLICK TO ACTIVATE THE GIMBAL WHILE SEATED (AFTER COMPLETING THE GUNS SETUP & E2)
b	AMMO BOXES TO POD CONTROLLER (ACTIVE → ACTIVE)	-----	-----
c	POD CONTROLLER TO POD CONTROLLER (CROSSHAIRS → ACTIVE)	-----	ENABLING CROSSHAIR
B	WIRING ACE GUNS (3 DIFFERENT WAYS TO FIRE)	-----	-----
d	OPTIONAL 1: WIRE THE ACE GUNS TO THE POD CONTROLLER E.G ACE GUN (FIRE) → POD CONTROLLER (MOUSE 1)	-----	E.G DIFFERENT KEYS FROM POD TO FIRE 37MM CANNON GUN (MOUSE 1) HEAVY MACHINE GUN 20MM (SHIFT)
e	OPTIONAL 2: WIRE THE ACE GUNS TO THE E2 CHIP E.G ACE GUNS (FIRE) → E2 (GUNS_SPLITTER)	-----	MOUSE AIM BASIC V1 E2 CHIP ACTS LIKE A USB SPLITTER, ALLOWING A SINGLE E2 CHIP TO CONTROL AND FIRE MULTIPLE ACE GUNS WITH ONE LEFT CLICK.
f	OPTIONAL 3: WIRE THE E2 TO THE ACE GUN E.G E2 (GUN_WL) → ACE GUN (WIRELINK)	-----	PLUG & PLAY (ONLY ONE SINGLE GUN) *LEFT CLICK TO FIRE
140	GIMBAL ISSUES?	-----	-----
A	GIMBAL NOT POINTING EXACTLY AT THE CROSSHAIR	-----	SOLUTION: TYPE /Zero in game chat for auto calibration
B	IF THE GIMBAL AIMS IN THE OPPOSITE DIRECTION WHEN ACTIVATED: NEGATE THE EYE VECTOR (LINE 119 E2 CODE) VIDEO LINK: EASY SOLUTION	-----	LIKELY CAUSED BY THE GIMBAL NOT BEING ALIGNED WITH THE MAP'S NORTH DIRECTION DURING SETUP (DARK/NAVY LONG LINE ON THE GIMBAL), WHICH I RECENTLY DISCOVERED. PIC #140 (WRONG), PIC #145 (CORRECT)
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fun fact: making this kind of documentation takes a lot of effort and is soul draining...	SUPPORT MY WORK ON YOUTUBE , GITHUB , OR KO-FI — YOUR SUPPORT SHOWS MY NICHE SKILLS AND GMOD R&D PROJECTS ARE APPRECIATED!	DISCORDS FOR ACF/ACE & WIREMOOD/STARFALL	CHECK OUT THEIR HOMEPAGE/GITHUB OR CLICK ONE OF BLUE LINK (HYPERLINKS)

Picture Board

#140

Explanation For #140B

#145

Explanation For #140B

MY NEXT PROJECT?!?!
Anime: Heavy Object
Genre: Sc-Fi

TO Engineer